



# LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



## Iranian-Affiliated PLC Targeting Requires Airport and Aerospace OT Remote-Access Review

Threat Report

|                       |                 |
|-----------------------|-----------------|
| <b>Classification</b> | TLP:CLEAR       |
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TLP:CLEAR | Lost Boys Cyber | Iranian-Affiliated PLC Targeting Requires Airport and Aerospace OT Remote-Access Review - Threat Report

Published: 2026-07-27 | TLP:CLEAR | Version: 1.0 | Author: Lost Boys Cyber

AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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# Iranian-Affiliated PLC Targeting Requires Airport and Aerospace OT Remote-Access Review

## 1. Executive Summary

CISA reporting on Iranian-affiliated actors accessing internet-facing PLCs is directly relevant to aviation OT environments where baggage handling, building automation, fuel systems, or manufacturing cells depend on PLCs and remote vendor access.

This report was selected for the 2026-07-27 daily Lost Boys Cyber production run. Source depth was sufficient for a defender-priority report, but any victim-specific details, exploitability claims, or attribution beyond the cited sources should be validated before operational use.

## 2. Intelligence Requirements

| Question                 | Current Answer                                                                                          | Confidence |
|--------------------------|---------------------------------------------------------------------------------------------------------|------------|
| Why prioritize now?      | Recent or materially updated public reporting places this issue in the daily watch window.              | High       |
| Who should review first? | airport OT, aerospace manufacturing, facility automation defenders and asset owners.                    | High       |
| What telemetry matters?  | Identity, endpoint process, network, SaaS, edge, mail, OT, and supplier-access telemetry as applicable. | Medium     |

## 3. Source Review & Confidence

| #                             | Source                                         | URL                                                                                                                                                               | Contribution                                        |
|-------------------------------|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 1                             | CISA AA26-097A Iranian-affiliated PLC activity | <a href="https://www.cisa.gov/news-events/cybersecurity-advisories/aa26-097a">https://www.cisa.gov/news-events/cybersecurity-advisories/aa26-097a</a>             | Supports selection, confidence, or defender action. |
| 2                             | CISA ICS recommended practices                 | <a href="https://www.cisa.gov/resources-tools/resources/ics-recommended-practices">https://www.cisa.gov/resources-tools/resources/ics-recommended-practices</a>   | Supports selection, confidence, or defender action. |
| 3                             | ENISA transport threat landscape               | <a href="https://www.enisa.europa.eu/publications/enisa-transport-threat-landscape">https://www.enisa.europa.eu/publications/enisa-transport-threat-landscape</a> | Supports selection, confidence, or defender action. |
| Source-Reliability Assessment |                                                | Value                                                                                                                                                             |                                                     |
| Overall confidence            |                                                | High                                                                                                                                                              |                                                     |
| Source window                 |                                                | Preferred 24-72 hours; expanded to 7 days or authoritative standing guidance where fewer current sector-specific sources were available.                          |                                                     |
| Evidence limits               |                                                | This report does not fabricate IOCs, exploit code, victim counts, or actor attribution not present in public sources.                                             |                                                     |

## 4. Threat / Vulnerability Details

CISA reporting on Iranian-affiliated actors accessing internet-facing PLCs is directly relevant to aviation OT environments where baggage handling, building automation, fuel systems, or manufacturing cells depend on PLCs and remote vendor access. The defensive focus is exposure validation, compensating control review, and near-term hunt coverage rather than speculative attribution.

## 5. Attack Scenarios

| Scenario                 | Preconditions                                                                                                | Defensive Review                                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Internet-facing exposure | Affected service, gateway, mail, browser, SaaS, OT, or supplier endpoint is reachable or weakly segmented.   | Confirm asset inventory, patch state, WAF/edge controls, and recent authentication anomalies.                   |
| Trusted-channel abuse    | Adversary uses a browser, updater, remote-access tool, SaaS integration, supplier account, or mail workflow. | Review child-process activity, consent grants, remote sessions, new services, and unusual data movement.        |
| Post-compromise staging  | Operator or malware performs discovery, credential access, archive staging, or exfiltration.                 | Hunt command shells, archive tools, credential-store access, unusual outbound connections, and new persistence. |

## 6. Aviation / Aerospace Sector Impact

| Segment                     | Operational Relevance                                                                                                         | Likely Exposure Paths                                                                             | Defender Actions                                                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Airlines / airports         | Potential disruption to passenger processing, corporate IT, OT support systems, or supplier operations.                       | Vendor remote access, identity compromise, SaaS integrations, exposed edge or OT systems.         | Validate MFA, segmentation, incident fallbacks, vendor contacts, and sector-specific telemetry.                |
| Aerospace / MRO / satellite | Potential impact to engineering, regulated maintenance, manufacturing continuity, ground segment, or sensitive supplier data. | Engineering workstations, PLM/MRO portals, remote administration, update channels, cloud tenants. | Review third-party assurance, privileged access, update integrity, backup isolation, and notification clauses. |

## 7. Detection Engineering

| Signal                                                                                                                                                                                                                                              | Telemetry Source               | Analytic Goal                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------|
| Rare scripting or LOLBin child process from browser/mail/remote-access/updater/service parent                                                                                                                                                       | EDR process creation           | Surface exploitation, phishing execution, trusted-channel abuse, or operator activity. |
| New external connection burst from scripting, browser helper, remote admin, or OT engineering workstation                                                                                                                                           | Network/proxy/EDR network      | Identify C2, staging, exfiltration, or unexpected supplier access.                     |
| New service, scheduled task, local admin group change, OAuth grant, or MFA reset                                                                                                                                                                    | Endpoint, identity, SaaS audit | Surface persistence and privilege changes.                                             |
| <p>title: LBC Daily Priority Suspicious Trusted Channel Execution<br/> id: lbc-tr-2026-154-trusted-channel-exec<br/> status: experimental<br/> description: Detects suspicious script/LOLBin execution from parent processes commonly abused in</p> |                                |                                                                                        |

```

vulnerability exploitation, phishing, browser abuse, supplier tooling, or remote administration.
logsource:
  product: windows
  category: process_creation
detection:
  parent_selection:
    ParentImage|contains:
      - '\\chrome.exe'
      - '\\msedge.exe'
      - '\\outlook.exe'
      - '\\zmmailbox'
      - '\\updater'
      - '\\vpn'
      - '\\thin'
      - '\\remote'
  child_selection:
    Image|endswith:
      - '\\powershell.exe'
      - '\\cmd.exe'
      - '\\wscript.exe'
      - '\\rundll32.exe'
      - '\\mshta.exe'
      - '\\curl.exe'
      - '\\7z.exe'
  condition: parent_selection and child_selection
level: high

```

## 8. Threat Hunting

### 8.1. Hypothesis 1: suspicious execution from trusted user-facing or supplier-managed tooling

```

DeviceProcessEvents
| where Timestamp > ago(14d)
| where InitiatingProcessFileName has_any
('chrome.exe', 'msedge.exe', 'outlook.exe', 'updater.exe', 'vpn.exe', 'remote.exe', 'thinclient.exe', 'java.exe')
| where FileName in~
('powershell.exe', 'cmd.exe', 'wscript.exe', 'rundll32.exe', 'mshta.exe', 'curl.exe', 'wget.exe', '7z.exe', 'python.exe')
| project Timestamp, DeviceName, InitiatingProcessFileName, FileName, ProcessCommandLine, AccountName

```

### 8.2. Hypothesis 2: abnormal outbound activity from scripting or remote-access context

```

DeviceNetworkEvents
| where Timestamp > ago(14d)
| where InitiatingProcessFileName has_any
('powershell.exe', 'cmd.exe', 'curl.exe', 'wget.exe', 'python.exe', 'chrome.exe', 'msedge.exe', 'vpn.exe', 'remote.exe')
| summarize connections=count(), first_seen=min(Timestamp), last_seen=max(Timestamp),
urls=make_set(RemoteUrl, 10) by DeviceName, InitiatingProcessFileName, RemoteIP, RemotePort
| where connections > 20 or array_length(urls) > 5

```

### 8.3. Hypothesis 3: identity or persistence changes after initial access

```

DeviceEvents
| where Timestamp > ago(14d)
| where ActionType has_any
('UserAccountAddedToLocalGroup', 'ScheduledTaskCreated', 'ServiceInstalled', 'OAuthAppConsentGranted', 'MfaRequirementSatisfied', 'TamperingAttempt')
| project Timestamp, DeviceName, ActionType, AccountName, AdditionalFields

```

## 9. MITRE ATT&CK Mapping

| Tactic         | Technique                                                | Rationale                                                                         |
|----------------|----------------------------------------------------------|-----------------------------------------------------------------------------------|
| Initial Access | T1190 Exploit Public-Facing Application / T1566 Phishing | Relevant to vulnerability exploitation, webmail phishing, browser abuse, supplier |

|                         |                                                             |                                                                                      |
|-------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------|
|                         |                                                             | exposure, or OT remote-access pathways.                                              |
| Execution               | T1059 Command and Scripting Interpreter                     | Common follow-on behavior for exploitation and operator-driven activity.             |
| Persistence / Privilege | T1053 Scheduled Task, T1543 Create or Modify System Process | Used as behavior-focused hunt coverage; validate locally.                            |
| Command and Control     | T1105 Ingress Tool Transfer / T1071 Web Protocols           | Covers covert browser/SaaS/HTTP C2 and staging paths.                                |
| Exfiltration            | T1567 Exfiltration Over Web Service                         | Relevant where reporting concerns SaaS, browser, ransomware, or data-theft activity. |

## 10. Recommended Actions

| Priority | Owner                   | Action                                                                                                                                 |
|----------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Critical | Asset owners            | Identify exposed products, services, browser/AI surfaces, OT remote-access systems, or supplier-managed paths relevant to this report. |
| High     | Security operations     | Run the included hunts over 14-30 days and triage hits against maintenance windows and known automation.                               |
| High     | Detection engineering   | Convert the Sigma and KQL into platform-native analytics with allowlists for approved administrative tools.                            |
| Medium   | GRC / vendor management | Confirm notification obligations, vendor MFA, and escalation contacts for supplier-dependent operations.                               |

## 11. References

- [1] CISA AA26-097A Iranian-affiliated PLC activity: <https://www.cisa.gov/news-events/cybersecurity-advisories/aa26-097a>
- [2] CISA ICS recommended practices: <https://www.cisa.gov/resources-tools/resources/ics-recommended-practices>
- [3] ENISA transport threat landscape: <https://www.enisa.europa.eu/publications/enisa-transport-threat-landscape>